

Student Portal System

Complete Setup Guide

DISTINCTION GUIDE

SESC Module · LBU

2025 / 2026



Software to Install

Software	Version	Notes
Java JDK (Temurin)	21 LTS	Tick Set JAVA_HOME + Add to PATH during install
IntelliJ IDEA Community	Latest	Install Lombok plugin after setup
Node.js	LTS	Download .msi from nodejs.org
MySQL + MySQL Workbench	8.0.45	Root password: root · Port: 3306
MongoDB Community	Latest	Install as Windows Service · Port: 27017
MongoDB Compass	Latest	GUI — connect to mongodb://localhost:27017
Docker Desktop	Latest	Engine must show Running before starting
Git	2.48.1+	
Postman	Latest	For API testing

CHOOSE HOW TO RUN THE PROJECT

B

Run with Docker (Much Easier)

One command starts everything — no manual setup needed

- 1 Install only Docker Desktop from docker.com — nothing else needed
- 2 Open a terminal (Command Prompt or PowerShell)
- 3 Navigate to your project folder:

```
D:
cd "enter your main folder path"
example: cd "student-portal-system"
```

- 4 Run the single command:

```
docker compose up --build -d
```

- 5 Open the portals — everything is running

Docker pulls MySQL, MongoDB and Node.js automatically — no installs, no IntelliJ, no manual config.

Portal	URL	Credentials
Student Portal	http://localhost:8080	Register a new account
Finance Portal	http://localhost:8082/portal	No login — enter invoice reference
Library Portal	http://localhost:8083/login	Student ID + PIN 000000

OR — OPTION A: MANUAL SETUP

A Run Manually

Install all software, follow the step-by-step guide below

1 Create MySQL Databases

Open **MySQL Workbench** → connect to *Local instance MySQL80* → click the SQL editor tab and run:

```
CREATE DATABASE IF NOT EXISTS studentdb;
CREATE DATABASE IF NOT EXISTS librarydb;
```

Click the  **lightning bolt** to execute. Right-click **Schemas** → **Refresh All** — both databases should appear.

No tables needed — Spring Boot creates them automatically on first run.

2 Connect MongoDB in Compass

- Open **MongoDB Compass**
- Click **+** next to CONNECTIONS
- Enter: `mongodb://localhost:27017` → **Save & Connect** → name it **SESC Portal**
- You should see admin, config, local — MongoDB is working

financedb is created automatically when the Finance service starts.

3 Run Finance Service (Node.js)

Open **Command Prompt** and navigate to your project folder:

```
D:
cd "enter your main folder path"
example: cd "student-portal-system\finance-service"
```

Install dependencies (only needed once):

```
npm install
```

Set environment variables and start:

```
set MONGO_URI=mongodb://localhost:27017/financedb
set PORT=8082
node src/app.js
```

You should see: Finance Service running on port 8082 — leave this terminal open.

4 Run Library Service in IntelliJ

Open **IntelliJ IDEA** → **File** → **Open** → navigate to your **library-service** folder inside your project directory:

```
D:
cd "enter your main folder path"
example: cd "student-portal-system\library-service"
```

Click **Trust Project** → wait for indexing → **Run** → **Edit Configurations** → **+** → **Spring Boot**

Set **Name:** Library Service | **Main class:** com.university.libraryservice.LibraryServiceApplication

Click **Modify options** → **Environment variables** → add:

Variable Name	Value
LIBRARY_DB_URL	jdbc:mysql://localhost:3306/librarydb?createDatabaseIfNotExist=true&serverTimezone=UTC
LIBRARY_DB_USER	root
LIBRARY_DB_PASSWORD	root
FINANCE_SERVICE_URL	http://localhost:8082

Click OK → green Run button → wait for: Started LibraryServiceApplication

5 Run Student Service in IntelliJ

File → **Open** → **New Window** → navigate to your **student-service** folder:

```
D:
cd "enter your main folder path"
example: cd "student-portal-system\student-service"
```

Trust Project → wait for indexing → **Run** → **Edit Configurations** → **+** → **Spring Boot**

Set **Name:** Student Service | **Main class:** com.university.studentservice.StudentServiceApplication

Click **Modify options** → **Environment variables** → add:

Variable Name	Value
STUDENT_DB_URL	jdbc:mysql://localhost:3306/studentdb?createDatabaseIfNotExist=true&serverTimezone=UTC
STUDENT_DB_USER	root
STUDENT_DB_PASSWORD	root
FINANCE_SERVICE_URL	http://localhost:8082

LIBRARY_SERVICE_URL	http://localhost:8083
FINANCE_PORTAL_URL	http://localhost:8082
LIBRARY_PORTAL_URL	http://localhost:8083

Click OK → green Run button → wait for: Started StudentServiceApplication

6 Open the Portals

Portal	URL	Credentials
Student Portal	http://localhost:8080	Register a new account on the login page
Finance Portal	http://localhost:8082/portal	No login — enter invoice reference number
Library Portal	http://localhost:8083/login	Student ID (e.g. STU-00001) + PIN 000000 Forced PIN change on first login

→ Startup Order — Always Follow This (Option A)

1	2	3	4	5
MySQL	MongoDB	Finance	Library	Student
Auto-starts as Windows service	Auto-starts as Windows service	CMD terminal node src/app.js	IntelliJ Run button	IntelliJ Always last

Student service must always start last — it depends on Finance and Library being ready.